Newton-Raphson Method

Exam code: 9MA0

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Written by: Lucy Kirkham

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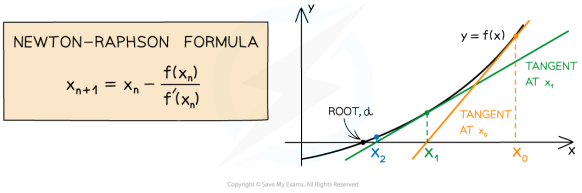
Newton-Raphson method

What is the Newton-Raphson method?

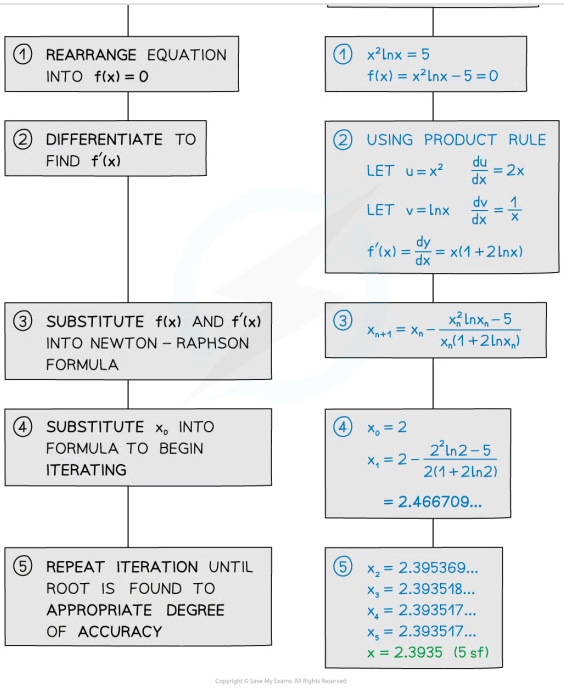
- The Newton-Raphson method finds **roots** of equations in the form **f(x) = 0**
- It can be used to find **approximate solutions** when an equation cannot be solved using the usual **analytical methods**
- It works by finding the **x-intercept** of **tangents to f(x)** to get closer and closer to a root



How do I use the Newton-Raphson method?

- After using **differentiation** to find f'(x) the formula uses iteration to come to an ever more accurate solution

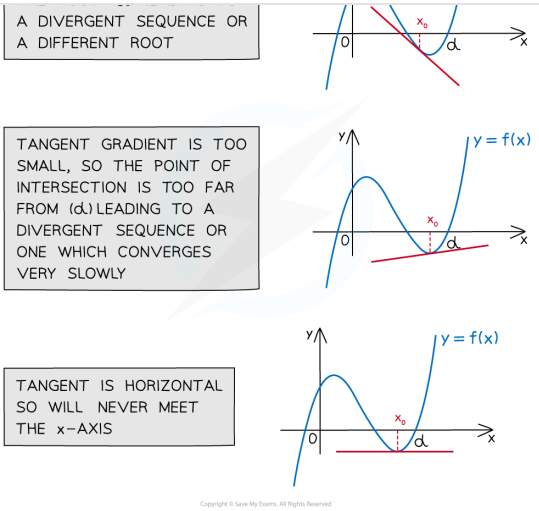
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Can the Newton-Raphson method fail?

The Newton-Raphson method can fail when:

- the starting value **x_0 is too far away** from the root leading to a **divergent** sequence or a **different root**
- the **tangent gradient** is **too small**, where f'(x) close to 0 leading to a **divergent** sequence or one which **converges very slowly**
- the **tangent** is **horizontal**, where f'(x) = 0 so the tangent will **never meet the x-axis**
- the equation **cannot be differentiated** (or is awkward and time-consuming to do)



Examiner Tips and Tricks

- The formula for the Newton-Raphson method is given in the formula booklet.
- Use ANS button on your calculator to calculate repeated iterations.
- Keep track of your iterations using x_2, x_3... notation.
- Newton-Raphson questions may be part of bigger numerical methods questions.

Worked Example

root of $x^3 - 2\sqrt{x^3} = 1$ to 4 decimal places, starting with $x_0 = 2$.

b) Explain why $x_0 = 1$ would have been an unsuitable starting value.

a) $x^3 - 2\sqrt{x^3} = 1$

STEP 1 REARRANGE EQUATION INTO $f(x) = 0$

$f(x) = x^3 - 2\sqrt{x^3} - 1 = 0$

$f(x) = x^3 - 2x^{\frac{3}{2}} - 1$

REWRITE AS POWERS TO MAKE DIFFERENTIATION EASIER

STEP 2 DIFFERENTIATE TO FIND $f'(x)$

$f'(x) = 3x^2 - 3x^{\frac{1}{2}}$

$= 3x^2 - 3\sqrt{x}$

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

STEP 3 SUBSTITUTE $f(x)$ AND $f'(x)$ INTO NEWTON – RAPHSON FORMULA

$x_{n+1} = x_n - \frac{x_n^3 - 2\sqrt{x_n^3} - 1}{3x_n^2 - 3\sqrt{x_n}}$

$x_0 = 2$

IT CAN BE EASIER TO WRITE x's FIRST THEN GO BACK AND ADD n's FOR x_n

STEP 4 SUBSTITUTE x_0 INTO FORMULA TO BEGIN ITERATING

$x_1 = 2 - \frac{2^3 - 2\sqrt{2^3} - 1}{3(2^2) - 3\sqrt{2}} = 1.82685...$

$x_2 = 1.80024...$

STEP 5 REPEAT ITERATION UNTIL ROOT IS FOUND TO REQUIRED DEGREE OF ACCURACY

$x_3 = 1.7996...$

$\vdots$

$x = 1.7996$ (4 dp)

USING "ANS" BUTTON

$ANS - \frac{ANS^3 - 2\sqrt{ANS^3} - 1}{3ANS^2 - 3\sqrt{ANS}}$

b) $f'(x) = 3x^2 - 3\sqrt{x}$

SUBSTITUTE $x_0 = 1$ INTO $f'(x)$

$f'(1) = 3(1)^2 - 3\sqrt{1} = 0$

$f'(1) = 0$ SO TANGENT WILL NEVER MEET x–AXIS

ie. THE NEWTON–RAPHSON METHOD WILL FAIL

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